Deformation invariance of plurigenera in fiberwise Kähler families

Tseng

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Abstract

We prove deformation invariance of every plurigenus in a proper surjective holomorphic submersion over a disk whose fibers are connected compact Kähler manifolds, in arbitrary dimension. From OpenAI’s logarithmic subadditivity and good divisorial decomposition results, we derive a finite ordinary minimal model program starting on a chosen smooth fiber with pseudoeffective canonical divisor. Hacon–Xie’s deformation theorem extends this program over the analytic family. Its semiample terminal central fiber is covered by the classical extension argument of Levine and Kollár–Mori. We also give a canonical extension theorem for central fibers that are smooth in codimension two and admit a compact Kähler modification. The proof combines Cohen–Macaulay reflexive canonical powers, logarithmic Hodge theory, and a Leray comparison to obtain a central period injection. Levine’s trace formula then annihilates the extension obstruction. Character summands recover every pluricanonical degree, and coherence converts finite-order lifts into analytic sections.

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1 Introduction and preliminaries

Deformation invariance of plurigenera for compact complex surfaces is due to Iitaka [10, Theorem III]. In arbitrary dimension, Siu established invariance for smooth projective families, first for fibers of general type [20, Main Theorem, p. 661] and subsequently in general [21, Corollary 0.3]. He also formulated the corresponding conjecture for compact Kähler manifolds [21, Conjecture 0.4]. We prove the following statement.

Theorem 1.1

Let $\Delta = \{t \in \mathbb{C} : |t| < 1\}$, let $\mathcal{X}$ be a complex manifold, and let $\pi : \mathcal{X} \to \Delta$ be a proper surjective holomorphic submersion. Suppose that every fiber X_t is connected, compact, and Kähler. Then, for every integer $m \geq 1$, the function

$$t \mapsto P_m(X_t) = \dim_{\mathbb{C}} H^0\left(X_t, K_{X_t}^{\otimes m}\right)$$

is constant on Δ.

Recent work has established invariance in several Kähler settings. Deng, Ning, Wang, and Zhou [4, Theorem 1.1] prove invariance when the total space is Kähler and every fiber has nef canonical bundle. Chen, Rao, and Wang [3, Theorems 1.2–1.4] obtain local invariance for smooth proper Kähler morphisms from the nefness of the central canonical bundle. Li, Păun, Wang, and Zhou [14, Theorem 1.2] prove invariance in arbitrary dimension for smooth proper families with Kähler total space, without a nefness assumption. Liu and Wan [16, Theorem 1.1] establish the corresponding logarithmic extension theorem for rational boundaries with relative simple normal crossings and coefficients in $[0, 1)$, again with Kähler total space.

The threefold case is also studied by Hacon, Li, and Rao [7, Theorem 6.5]. Feng, Li, and Rao [5, Theorems 1.4 and 1.5] prove invariance for $\mathbb{Q}$-factorial Kähler families of threefolds with canonical central fiber and for smooth Kähler families whose central fiber has maximal Albanese dimension. In their terminology, a Kähler family has Kähler total space.

Theorem 1.1 establishes invariance in arbitrary dimension under the fiberwise Kähler hypothesis. A related local question, with nef canonical bundle on the central fiber, is posed in [3, Question 1.6]. The cases of fiber dimension at most two are classical and are recalled in Section 5. In higher dimension, the argument is local on the base and is carried out separately for each pluricanonical degree. Upper semicontinuity settles the case of a vanishing central plurigenus. Otherwise, a nonzero pluricanonical section makes the central canonical divisor pseudoeffective, and the proof proceeds through three constructions.

First, OpenAI’s logarithmic subadditivity theorem [18, Corollary 6.2] supplies the reduced simple normal crossing case of logarithmic Iitaka subadditivity assumed in [19, Assumption 1.1]. We therefore use the good divisorial decomposition theorem

[19, Theorem 2.13] and the ordinary and relative model constructions of that paper. Proposition 2.1, proved in Section 2, derives from these constructions a finite ordinary K-MMP beginning with the chosen smooth fiber and ending at a terminal model with semiample canonical divisor. Its proof uses the good decomposition in the ambient dimension and adapts the fixed-part, small-model, and nef-interval constructions of [19, Proposition 3.8 and Lemmas 3.11–3.12], together with the comparison of negative multiplicities in the proof of [19, Theorem 3.13].

Second, the deformation step is supplied by Hacon and Xie [8, Theorem 7.7]. Their theorem extends prescribed central MMP steps in proper flat analytic families over a smooth curve with Kähler canonical central fiber. Its Kähler hypothesis concerns the central fiber; in our application the boundary and generalized nef data are zero, so the additional boundary conditions are automatic. Proposition 3.2 gives the ordinary, relatively projective construction explicitly, including the central canonical algebra and the comparison of pluricanonical sections in every degree. The proof follows Kollár's deformation framework [11, paragraph 8], using Nakayama's analytic restriction and deformation results [17, VI.3.8 and VI.5.2].

Finally, the semiample terminal extension result needed at the endpoint is classical: the proof of Kollár–Mori [12, Theorem 12.5.5] applies in Fujiki's class $\mathcal{C}$, as recorded by Kollár [11, comments following Conjecture 17]. Theorem 4.1 extends this terminal class-$\mathcal{C}$ formulation to canonical central fibers smooth in codimension two and admitting a compact Kähler modification. We prove this canonical statement using the cyclic-cover method of Levine [13, Proposition 1.6 and Corollaries 1.8 and 1.10] and Kollár–Mori. Cohen–Macaulayness identifies the relevant first cohomology with that of the smooth locus. On a compact Kähler resolution of the central cover, Deligne's logarithmic Hodge-to-de Rham degeneration, in the formulation of Liu, Rao, and Wan [15, Theorem 3.2], gives an injection into logarithmic de Rham cohomology. A Leray comparison and de Rham naturality then imply that the central period map is injective. Its infinitesimal counterpart annihilates the extension obstruction. The character summands recover every lower pluricanonical degree, and coherence converts finite-order lifts into analytic sections. Birational invariance completes the proof in Section 5.

1.1 Notation and conventions

The letter n denotes the dimension of the fibers and their bimeromorphic models. For a normal $\mathbb{Q}$-Gorenstein space, $\omega_X^{[m]}$ denotes the reflexive canonical power. The notation $L \sim_{\mathbb{Q}} L'$ means that there is a positive integer r for which rL and rL' are holomorphic line bundles and are isomorphic. Semiampleness means global generation of a positive Cartier multiple.

A normal space is *globally strongly $\mathbb{Q}$-factorial* if every global coherent rank-one reflexive sheaf has an invertible reflexive power. In particular, every global Weil divisor has a positive Cartier multiple. Here the reflexive sheaves are taken on the whole space. A bimeromorphic map is *small* if it is an isomorphism in codimension one. An ordinary *K-minimal model program*, or *K-MMP*, consists of divisorial

contractions and flips of K-negative extremal rays. Braces $\{D\}$ denote the Bott–Chern class of a divisor. In formulas involving classes, a rational line bundle is identified with its first Chern class. Canonical comparisons are understood through local meromorphic canonical identifications; their discrepancy divisors are global.

Following Boucksom's analytic divisorial Zariski decomposition, we write $N(\alpha)$ for the divisorial negative part of a pseudoeffective Bott–Chern class. On a smooth compact Kähler model, with Kähler form ω, the divisorial minimal multiplicities are

$$\nu_D(\alpha) = \lim_{\epsilon \downarrow 0} \inf_{T \in \alpha,\, T \geq -\epsilon\omega} \nu_D(T), \qquad N(\alpha) = \sum_D \nu_D(\alpha) D.$$

The support of this sum is finite [1, Theorem 3.12(i)]. The class is *modified nef* when all its minimal multiplicities along prime divisors on the current space vanish. On normal models, we use these notions on smooth resolutions and keep track of all divisorial places. Comparisons between models are made on a common smooth resolution.

1.2 Birational results for compact Kähler spaces

We use the following consequences of [18; 19]. The statements concern compact Kähler spaces and the indicated proper or projective morphisms.

(B1) Good canonical decomposition. If X is smooth compact Kähler and K_X is pseudoeffective, there is a smooth compact Kähler modification $\mu : Y \to X$ such that

$$\mu^* K_X \sim_{\mathbb{Q}} P + R, \qquad P \text{ is semiample}, \qquad R = N\left(\mu^* K_X\right) \geq 0, \tag{1}$$

where R is a rational divisor. On every higher smooth modification a, one has $N\left(a^*\mu^* K_X\right) = a^* R$. This is [19, Theorem 2.13 and Lemma 2.4].

The morphism μ may be chosen projective. Indeed, apply analytic flattening [9] to the original modification. After a sequence of blowups of X, its main transform is flat. It is proper of relative dimension zero and degree one, hence finite flat of degree one. The associated rank-one unital algebra is $\mathcal{O}$, so the main transform is an isomorphism. A projective resolution then gives a projective modification of X dominating the original one. Pullback preserves (1) and the asserted negative-part identities.

(B2) Ordinary steps and finiteness at positive scaling parameters. On a globally strongly $\mathbb{Q}$-factorial compact Kähler terminal space, we use the ordinary scaling programs constructed in [19, Lemma 3.6]. Their steps are projective contractions and flips of K-negative extremal rays. The resulting spaces remain compact Kähler, terminal, and globally strongly $\mathbb{Q}$-factorial. The rays are tested against all Bott–Chern classes. The cone theorem recalled in [8, Theorem 1.3] gives the rational-curve length bound $2n$. Following [19, Section 3.1], we use the uniform bound $N_n = 2n + 1$.

Starting with a Kähler scaling class, such a K-program can be run, and only finitely many steps occur in every parameter interval $[\epsilon, T]$ with $\epsilon > 0$. If K is pseudoeffective and the program does not reach a nef model after finitely many steps, its positive scaling parameters tend to zero. The corresponding statements apply to $K + bP$, where P is a nef rational line bundle on the current model: if mP is Cartier and $b \in \mathbb{Q}$ satisfies $b > 4nm$, the steps can be chosen K -negative and P-trivial.

The effective comparisons are those of [19, Lemma 3.7]. The projective construction of the steps and finiteness at positive scaling parameters are supplied by [19, Proposition 3.42 and Corollary 3.50].

(B3) Fixed Cartier indices and transport of classes. If a Cartier line bundle mP has degree zero on a contraction as above, it descends as a Cartier line bundle to the target, with the same integer m. Nefness and global generation of a fixed multiple are preserved. A real Bott–Chern class of degree zero on the contracted ray also descends to the contraction target. See [19, Lemmas 3.3 and 3.15].

The fixed-index assertion can also be seen from relative klt base-point-freeness: two consecutive sufficiently large powers descend, and their quotient descends the original line bundle. The same Cartier exponent is therefore available throughout the program.

(B4) Big relative good models. Let M be compact Kähler, terminal, and globally strongly $\mathbb{Q}$-factorial, with semiample K_M. Let $g : M \to Z$ be its associated morphism with connected fibers, where Z is normal projective. Let A_M be modified nef and big, represented by globally nef generalized data on a fixed smooth carrier. If $L = K_M + tA_M$, for some $t > 0$, defines a generalized klt pair, then L has a nonextracting good log terminal model over Z. The output is again globally strongly $\mathbb{Q}$-factorial and generalized klt. Its comparison error is effective, exceptional over the output, and strictly positive along each contracted prime divisor of the source. The construction preserves the morphism to Z and the forward Bott–Chern traces.

Here L is globally big, since K_M is nef and tA_M is big. We use the proper-relative good-model assertion of [19, Proposition 3.2(ii) and Theorem 3.49], with its globally big construction retaining the chosen base [19, Lemma 3.45]. In Lemma 2.4, that base is the image of the semiample morphism defined by K_M; properness of this morphism is sufficient. The generalized klt adjoints occurring here satisfy the same uniform rational-curve length bound N_n.

(B5) Projective birational relative programs. Given a projective bimeromorphic morphism from a smooth compact Kähler space to a normal compact Kähler target, the ordinary relative K-program can be run and terminated. Its terminal output remains projective over that target and globally strongly $\mathbb{Q}$-factorial. See [19, Proposition 3.4].

1.3 Divisorial negative parts

We use exceptional translation on resolutions of normal Kähler spaces in the form of [19, Lemma 2.3]. The subtraction and fixed-section properties are recalled in [19, Lemmas 2.2 and 2.5]. If $q : W \to T$ is a modification, A is pseudoeffective, and $E \geq 0$ is q-exceptional, then

$$N\left(q^{*}A + E\right) = N\left(q^{*}A\right) + E. \tag{2}$$

If $0 \leq F \leq N\left(A\right)$, then

$$N\left(A - F\right) = N\left(A\right) - F. \tag{3}$$

Nef classes have zero negative part, addition of a nef class does not increase divisorial minimal multiplicities, and strict-transform multiplicities are unchanged under modifications. The divisor of any nonzero section of a line bundle contains the corresponding negative part.

We also use the following nef-direction form of the continuity argument in [19, Lemma 3.7]. If A is pseudoeffective and B is nef, then, for every fixed divisorial place Q,

$$\nu_{Q}\left(A + \epsilon B\right) \to \nu_{Q}\left(A\right) \qquad (\epsilon \downarrow 0). \tag{4}$$

To see this, work on a smooth model carrying Q, choose a Kähler class Ω, and choose c sufficiently large that $c\Omega - B$ is nef. Then

$$\nu_{Q}\left(A + c\epsilon\Omega\right) \leq \nu_{Q}\left(A + \epsilon B\right) \leq \nu_{Q}\left(A\right).$$

The left side tends to the right side by the definition of minimal multiplicities. This proves (4). By homogeneity, the same observation applies to $\left(1 - s\right)A + sB$.

Finally, an exceptional real divisor with zero Bott–Chern class is zero. Both it and its negative are relatively nef, so the negativity lemma applies to each sign. We shall use this observation to recover exceptional-divisor identities from class comparisons.

2 Finite minimal model programs on the original space

We derive a finite ordinary K_{X}-MMP beginning with the given manifold X from the birational constructions of [19]. This form of the conclusion supplies the individual contractions and flips to be extended over the family in Section 3. The Kodaira-dimension-zero case is contained in [19, Corollary 3.10].

OpenAI’s special-termination theorem [19, Theorem 3.13] asserts that the exceptional, flipping, and flipped loci of a chosen dlt scaling eventually avoid the supports of the boundary floors on the successive models. For zero boundary, that

condition places no restriction on the remaining flips. The proposition below establishes finiteness of a chosen entire ordinary K-program, using the good decomposition [19, Theorem 2.13] in the ambient dimension.

Proposition 2.1

Let X be a smooth compact Kähler manifold with K_X pseudoeffective. Then there is a finite ordinary K_X-MMP whose endpoint $X_{\min}$ is a compact Kähler terminal space with semiample canonical divisor $K_{X_{\min}}$.

The proof reduces a potentially infinite scaling to an ambient terminal model T with $N(K_T) = 0$, and restarts with a Kähler scaling direction. A small comparison model then makes the transforms of $K_T + sA$ nef on one interval containing $s = 0$. On a common resolution, every divisorial negative multiplicity is consequently affine throughout that interval. The comparison in the proof of [19, Theorem 3.13] shows that a nontrivial block of negative flips at a scaling wall changes the slope of at least one such multiplicity. This forces a restarted program on T to terminate; adjoining the finite initial segment gives the required program on X.

2.1 Contraction of the fixed part

The following is the terminal, boundary-free case of OpenAI's fixed-part contraction theorem [19, Proposition 3.8]. We recall the argument with the scaling comparison that will also be used in the final step.

Lemma 2.2

Let T be a compact Kähler terminal space that is globally strongly $\mathbb{Q}$-factorial, and suppose that

$$K_T \sim_{\mathbb{Q}} P_T + D_T, \qquad P_T \ \text{ nef}, \qquad D_T \geq 0,$$

where the terms are rational line bundles or rational divisors, and $N\left(p^{*}K_T\right) = p^{*}D_T$ on every smooth resolution p. Then there is a finite K_T-negative, P_T-trivial ordinary MMP whose endpoint satisfies $K \sim_{\mathbb{Q}} P$. If P_T is semiample, then the canonical divisor of the endpoint is semiample.

Proof

Choose m such that mP_T is Cartier, and then choose an integer $b > \max\{4nm, mN_n\}$. Put $H = K_T + bP_T$ and choose a Kähler class A with $H + A$ nef. By (B2), run the H-MMP with scaling of A, and denote its successive forward traces by A_i. An H-negative ray generated by C is also K-negative, and its generator may be chosen so that $0 < -K{\cdot}C \leq N_n$. If $P{\cdot}C > 0$, the fixed Cartier index gives $P{\cdot}C \geq 1/m$, whence $H{\cdot}C \geq -N_n + b/m > 0$, a contradiction. Thus every step is P-trivial. Statement (B3) preserves both the same index m and nefness.

Let T_i be the ith working model, with $T_0 = T$, set $\lambda_{-1} = 1$, and let λ_i be its next scaling threshold. Then $H_i + \lambda A_i$ is nef for $\lambda \in [\lambda_i, \lambda_{i-1}]$; consecutive thresholds may coincide. At a step with $\lambda_i > 0$, let $E_i \geq 0$ be the H-comparison divisor on a common model. The class at the threshold is pulled back from the contraction base, so its two pullbacks agree. Consequently, for $0 \leq \lambda \leq \lambda_i$,

$$p_i^*(H_i + \lambda A_i) - q_i^*(H_{i+1} + \lambda A_{i+1}) = \left(1 - \frac{\lambda}{\lambda_i}\right)\{E_i\}. \tag{5}$$

For a finite initial segment and a parameter $\lambda \geq 0$ no larger than any threshold crossed in that segment, summing these comparisons gives an effective error divisor exceptional over the output. If the corresponding class on the output is nef, (2) identifies this error with the entire divisorial negative part of the original class on the common model. We use this formula below for $H = K + bP$, and later for $H = K$.

On the initial model and on every higher model, (3) gives

$$H \sim_{\mathbb{Q}} (1+b)K_T - bD_T, \qquad N(H) = (1+b)D_T - bD_T = D_T.$$

The descent of the line bundle P and the canonical comparison preserve this identity along the MMP: on a common model one has $p^*D = q^*D_i + E$, and (2) gives the same formula for the negative part. The prime divisor contracted by a divisorial step is covered by contracted curves. Since P has degree zero on those curves, $D{\cdot}C = K{\cdot}C < 0$. An effective rational Cartier divisor has nonnegative degree on every curve not contained in its support, as follows after clearing denominators. Hence the contracted prime divisor belongs to $\operatorname{Supp} D$. If the MMP terminates at a model on which H_i is nef, then $D_i = N(H_i) = 0$, so $K_i \sim_{\mathbb{Q}} P_i$.

It remains to exclude an infinite MMP. In that case its positive scaling parameters tend to zero. By (4), each component of the original D_T has positive coefficient in $N(H+\lambda A)$ for all sufficiently small positive λ. On a nef model for such a parameter, the negative part is entirely computed by the effective exceptional comparison divisor. If the component survived as a strict prime divisor on that model, its coefficient in the exceptional comparison would be zero, a contradiction. Thus each component of D_T is contracted at some finite stage. There are only finitely many components. After all these stages, $K \sim_{\mathbb{Q}} P$, so K is nef and the MMP terminates. The Cartier descents preserve a fixed globally generated multiple, so they also preserve semiampleness.

□

2.2 Canonical decompositions on intermediate models

Fix (1). Let T be a terminal model obtained from X by a finite ordinary MMP. Choose a smooth common model W dominating Y, X and T, and write $r : W \to X$ and $s : W \to T$ for the resulting morphisms. The canonical meromorphic comparisons give

$$K_W = r^*K_X + E_X = s^*K_T + E_T, \qquad E_X, E_T \geq 0,$$

where the error divisors are exceptional for the respective morphisms. The finite K-negative program has the canonical comparison $r^*K_X = s^*K_T + G$, where $G \geq 0$ is s-exceptional. If Θ is a positive current in $c_1(K_X)$, then $s_*r^*\Theta$ is a positive current in $c_1(K_T)$, since $s_*[G] = 0$. Hence K_T is pseudoeffective. Pulling back (1) and applying (2), we obtain

$$K_W \sim_{\mathbb{Q}} P_W + N_W, \qquad P_W \ \text{ semiample}, \qquad N_W = N(K_W).$$

A second application of (2) gives

$$R_T := N_W - E_T = N\left(s^*K_T\right) \geq 0, \qquad s^*K_T \sim_{\mathbb{Q}} P_W + R_T. \tag{6}$$

Thus the initial good decomposition induces (6) on every finite intermediate model.

2.3 A small semiample comparison model

We adapt the normalized-graph construction of [19, Lemma 3.11] to canonical divisors on terminal spaces.

Lemma 2.3

Suppose that the model T above satisfies $N(K_T) = 0$, with vanishing at every prime divisor on T. Then there is a globally strongly $\mathbb{Q}$-factorial compact Kähler terminal model M that is small bimeromorphic to T and has semiample canonical divisor K_M. The model M will serve as a comparison model for an ordinary K_T -MMP.

Proof

The divisor R_T in (6) is now s-exceptional. Choose m sufficiently divisible that mP_W is globally generated and mR_T is integral. The fixed-part property gives

$$H^0(T, mK_T) = H^0\left(W, ms^*K_T\right) = s_{mR_T} H^0(W, mP_W).$$

Let $\Gamma \to T$ be the normalization of the graph of this complete pluricanonical map. This morphism is projective, and Γ carries a semiample rational line bundle P_Γ whose pullback to a sufficiently high model over W is precisely P_W. Choose a resolution $W_0 \to \Gamma$ with W_0 smooth and the morphism projective, and apply statement (B5) to run an ordinary relative K-MMP over Γ. Write its endpoint as $g : Y_0 \to \Gamma$, and let $h : Y_0 \to T$ be the composite. The space Y_0 is terminal. Since T is terminal,

$$K_{Y_0} = h^*K_T + F,$$

where $F \geq 0$ is h-exceptional and has strictly positive coefficient at every h -exceptional prime divisor.

Set $P_0 = g^*P_\Gamma$. Pushing down (6) from a common model gives

$$h^*K_T \sim_{\mathbb{Q}} P_0 + R_0, \qquad R_0 \geq 0 \ \text{ is an } \ h\text{-exceptional rational divisor.}$$

Since Y_0 is globally strongly $\mathbb{Q}$-factorial, R_0 is rational Cartier, so the following pullbacks are defined. On a common model $a : W' \to W$, $v : W' \to Y_0$, the divisor $a^*R_T - v^*R_0$ is v-exceptional and rationally linearly equivalent to zero. Applying negativity with both signs shows that it vanishes. Therefore

$$K_{Y_0} \sim_{\mathbb{Q}} P_0 + (R_0 + F), \qquad N\left(v^*K_{Y_0}\right) = v^*(R_0 + F).$$

The second identity also follows directly by applying (2) to hv.

Lemma 2.2 contracts $R_0 + F$. Its support contains every h-exceptional prime divisor and contains no prime divisor coming from T. Moreover, every divisor contracted by the MMP belongs to this support. The resulting model M is therefore small bimeromorphic to T, terminal and Kähler, with $K_M \sim_{\mathbb{Q}} P_M$ semiample. It is globally strongly $\mathbb{Q}$-factorial by statements (B5) and (B2).

□

2.4 A common nef model for an interval

The construction of [19, Lemma 3.12] produces one model that is nef for an entire interval of adjoints. We record its exceptional comparison and the resulting affine formula for the negative part, as used in the proof of [19, Theorem 3.13].

Lemma 2.4

In the setting of Lemma 2.3, let A be a Kähler class on T. There are a globally strongly $\mathbb{Q}$-factorial compact Kähler terminal model V small bimeromorphic to T, a forward trace A_V of this class, and $\delta > 0$ such that

$$K_V + sA_V \ \text{ is nef} \qquad (0 \le s \le \delta).$$

On a common model $p : U \to T$, $q : U \to V$, there is an effective real divisor $F(s)$, exceptional for both morphisms and with coefficients affine in s, such that

$$\begin{aligned} p^*(K_T + sA) &= q^*(K_V + sA_V) + \{F(s)\}, \\ N(p^*(K_T + sA)) &= F(s) \end{aligned} \tag{7}$$

throughout $[0, \delta]$. This affine dependence persists on every higher smooth model.

Proof

First transport h^*A to M along the finite construction of Lemma 2.3. At each K_i-negative step, choose a generating curve C_i and set $c_i = (\alpha_i \cdot C_i)/(K_i \cdot C_i)$. The class $\alpha_i - c_i K_i$ vanishes on the entire analytic numerical ray. By statement (B3), it descends to the contraction base. On the next model, add back $c_i K_{i+1}$. The difference on a common model is c_i times the canonical comparison divisor. Summing finitely many such comparisons produces A_M. On a common model $p : U \to T$, $q : U \to M$,

$$q^*A_M - p^*A = \{F_A\}, \tag{8}$$

where F_A is exceptional for both morphisms. The divisor $-F_A$ is q-nef, since it has the same degree as p^*A on every q-contracted curve. Negativity gives $F_A \ge 0$. Applying (2) to p yields $N\left(q^*A_M\right) = F_A$, so A_M is modified nef. It is also big, since p^*A is big.

Set $P_M = K_M$. On the fixed common carrier, take tp^*A as the globally nef b-data, and take zero ordinary boundary on M. This gives the generalized adjoint $L = P_M + tA_M$. On a fixed log resolution, its boundary is the ordinary crepant boundary plus tF_A. Since M is terminal, all coefficients remain strictly less than one for a sufficiently small fixed $t > 0$, so the pair is generalized klt. The total

boundary-plus-nef trace is globally modified big, and L is globally big and modified nef.

Write $mP_M = g_M^* H$, where $g_M : M \to Z$ has connected fibers, Z is normal projective, and H is ample Cartier. Applying statement (B4) gives a globally strongly $\mathbb{Q}$-factorial good model $M \dashrightarrow V$ over Z. Set $P_V = g_V^* H/m$. This map is small. Indeed, on a common model one has $a^*L = c^*L_V + E$, with $E \geq 0$ and with strictly positive coefficient at every original prime divisor that is contracted. The class L_V is a pseudoeffective forward trace, so (2) gives $N\left(a^*L\right) \geq E$. Since L is modified nef, its negative part has coefficient zero at every original prime divisor. This excludes all divisorial contractions; nonextraction excludes divisors in the other direction.

The map is over Z and small, so $K_V \sim_{\mathbb{Q}} P_V$. On a common model, the difference of the two canonical pullbacks is exceptional and has zero class, hence vanishes. Thus V is still terminal. Define $A_V = (L_V - P_V)/t$. Choose $b > mN_n$. Then $L_V + bP_V$ is globally nef. Otherwise a negative ray has a generating curve C with $0 < -L_V \cdot C \leq N_n$. If $P_V \cdot C = 0$, then C is vertical over Z, contrary to the relative nefness of L_V. If $P_V \cdot C > 0$, then $P_V \cdot C \geq 1/m$, again contradicting $b > mN_n$.

Both P_V and $(1+b)P_V + tA_V$ are nef. Convexity therefore gives

$$P_V + sA_V \quad \text{is nef}, \qquad 0 \leq s \leq \delta = \frac{t}{1+b}.$$

Pull all comparisons to a common smooth model of T and V. Since the bimeromorphic map is small, the difference $F(s)$ is exceptional for both morphisms, and the linear transport makes its coefficients affine in s. The divisor $-F(s)$ is p-nef because $q^*(K_V + sA_V)$ is nef. Negativity gives $F(s) \geq 0$. Applying (2) to q proves (7). On a higher model, the pullback of the nef class remains nef and the pullback of the error remains exceptional. Hence the entire statement persists.

□

2.5 Termination of the MMP

Proof of Proposition 2.1

Starting from the original smooth manifold X, run the ordinary K_X-MMP with scaling supplied by statement (B2). If it reaches a nef endpoint after finitely many steps, proceed to the final paragraph of the proof. Suppose that it is infinite. Every original prime divisor that is contracted has positive coefficient in $N(K_X)$: its coefficient in the canonical comparison at the relevant finite stage is strictly positive, while the canonical class on the other side remains pseudoeffective. Formula (2) gives the asserted lower bound. The MMP is nonextracting, and $N(K_X)$ has finite support. Hence only finitely many divisorial steps occur.

After the last divisorial step, pass also beyond the current finite block of steps having the same threshold, and denote the resulting model by T. All subsequent steps are flips, and T has a nonempty interval of positive parameters on which it is a nef model. Denote the trace of the original scaling direction by H; it need not be nef. Choose $\lambda_0 > 0$ in that interval so that $B = K_T + \lambda_0 H$ is nef. For subsequent scaling parameters $\lambda > 0$ tending to zero, the small nef models give the exceptional comparisons for $K_T + \lambda H$. Its negative part therefore has coefficient zero at every prime divisor on T. Using

$$K_T + \lambda H = \left(1 - \frac{\lambda}{\lambda_0}\right) K_T + \frac{\lambda}{\lambda_0} B$$

and (4), we obtain $N(K_T) = 0$. The limit argument uses the nef class $B = K_T + \lambda_0 H$.

Restart the ordinary K_T-MMP with scaling of a Kähler class A on T. Since the negative part of K_T is zero, the new MMP contracts no divisor: the canonical comparison argument above would force a strictly positive generic multiplicity ν at any contracted prime divisor. Thus all its nontrivial steps are small. Lemma 2.4 supplies a fixed comparison model V and an interval $[0, \delta]$.

If the new MMP were infinite, its positive parameters would tend to zero. Every interval bounded away from zero contains only finitely many steps, so there would be infinitely many distinct positive thresholds. Combine the finitely many consecutive steps at each threshold s_i into a single block. On a common resolution of the models immediately before and after this block, let $D_i \geq 0$ be the canonical comparison divisor. Strict K-negativity of the steps gives $D_i \neq 0$. At the wall $s = s_i$, the pullbacks of the scaled classes agree. Thus, for every s,

$$p_i^*(K_i + sA_i) - q_i^*(K_{i+1} + sA_{i+1}) = \left(1 - \frac{s}{s_i}\right)\{D_i\}. \tag{9}$$

On each side of the wall, consecutive distinct thresholds bound a nonempty open interval. The corresponding nef models compute the entire negative part of the

original class $K_T + sA$. On a resolution simultaneously dominating these models and the fixed model V, choose a prime divisor whose coefficient d_i in D_i is positive. Equation (9) shows that the slopes of the affine formulas for this divisorial multiplicity on the two sides of the wall differ by $-d_i/s_i \neq 0$. Choose $0 < s_i < \delta$. This contradicts the *single* affine formula in (7) on the whole interval $[0, \delta]$. The new MMP is therefore finite. Concatenating it with the finite initial segment of the original MMP ending at T gives a finite MMP starting from the original X.
Let $X_{\min}$ be its endpoint, with $K_{X_{\min}}$ nef. On a resolution, (6) gives $s^*K_{X_{\min}} \sim_{\mathbb{Q}} P_W + N\left(s^*K_{X_{\min}}\right)$. Nefness makes the negative part vanish. Choose a positive integer q such that $qK_{X_{\min}}$ is Cartier and $s^*\mathcal{O}_{X_{\min}}\left(qK_{X_{\min}}\right)$ is globally generated. This is possible by taking a common multiple of the Cartier index downstairs and a generating multiple upstairs. Since $s_*\mathcal{O}_W = \mathcal{O}_{X_{\min}}$, the projection formula identifies their spaces of global sections. A base point downstairs would make its entire nonempty inverse image a base locus upstairs, a contradiction. Thus $K_{X_{\min}}$ is semiample. □

Remark

Strict K-negativity is used in (9) to obtain $D_i \neq 0$. The deformation argument below applies to the resulting finite ordinary K_X-MMP beginning with X.

3 Deformation of finite minimal model programs

Hacon and Xie [8, Theorem 7.7] prove that prescribed central MMP steps extend in proper flat analytic families over a smooth curve with Kähler canonical central fiber. Applied with zero boundary and zero generalized nef data, their theorem supplies the deformation step needed after Proposition 2.1. We give the ordinary, relatively projective construction in Proposition 3.2, identifying the central flip algebra in every degree and the pluricanonical section spaces on nearby fibers.

The argument follows the deformation framework of Kollár [11, paragraph 8]. Its form for generalized pairs with projective central fibers is developed by Hacon, Li, and Rao [7, Proposition 4.16], with the Kähler threefold case noted in [7, Remark 4.17]; Hacon–Xie treat Kähler central fibers in arbitrary dimension. See also [5, Theorem 2.28].

3.1 Birational invariance of reflexive pluricanonical sections

We begin with the standard birational invariance statement in the sheaf-theoretic form used to compare the fibers of the successive models.

Lemma 3.1

Let Z be a normal complex space with canonical singularities, and let $p : W \to Z$ be a proper resolution with W smooth. Under the usual identification of canonical differentials, one has

$$p_{*}\omega_{W}^{\otimes m} = \omega_{Z}^{[m]} \tag{10}$$

for every integer $m \geq 1$. Consequently, if Z_1 and Z_2 are bimeromorphic compact normal spaces with canonical singularities, then

$$H^{0}\left(Z_1, \omega_{Z_1}^{[m]}\right) \simeq H^{0}\left(Z_2, \omega_{Z_2}^{[m]}\right) \qquad (m \geq 1). \tag{11}$$

The identities hold for the reflexive canonical powers in every degree.

Proof

The asserted sheaf identity is local on Z. On each sufficiently small open subset of Z, choose a meromorphic canonical differential and use its pullback to define compatible local canonical divisors on W. Write

$$K_{W} = p^{*}K_{Z} + E, \qquad E \geq 0,$$

where E is a p-exceptional rational divisor. Let η denote the chosen meromorphic top differential. A local section s of the reflexive mth canonical power can be written as $s = u\eta^{\otimes m}$, with

$$D_{s} := \operatorname{div}(u) + mK_{Z} \geq 0.$$

Since K_{Z} is rational Cartier, D_{s} is an effective rational Cartier divisor. Its pullback is therefore effective. Viewed as a meromorphic m-canonical differential on W, the pullback of s has divisor

$$\operatorname{div}\left(p^{*}u\right) + mK_{W} = p^{*}D_{s} + mE \geq 0,$$

and is thus holomorphic. Conversely, a holomorphic m-canonical differential on W restricts to the common dense open set. It has no pole at any codimension-one point of Z, and hence extends uniquely as a section of the reflexive canonical power. This proves (10) on every open subset. Applying it on a common smooth resolution of Z_1 and Z_2 and taking global sections gives (11).

□

3.2 Deformation of a prescribed finite program

Proposition 3.2

Let $f : \mathfrak{X} \to D$ be a proper flat analytic family over a disk D containing the origin. Suppose that X_0 is connected, normal, compact, and has canonical singularities. Let

$$X_0 = X_0^{(0)} \dashrightarrow X_0^{(1)} \dashrightarrow \cdots \dashrightarrow X_0^{(r)} \tag{12}$$

be a prescribed finite ordinary K-MMP. Assume that every central contraction $\varphi_0^{(i)} : X_0^{(i)} \to Z_0^{(i)}$ is projective with connected fibers and that $-K_{X_0^{(i)}}$ is relatively ample. Assume also that every central output is normal with canonical singularities and, in the flipping case, that the prescribed morphism $X_0^{(i+1)} \to Z_0^{(i)}$ is the corresponding flip.

After shrinking D, there are proper flat analytic families $\mathfrak{X}^{(i)} \to D$ whose central fibers are precisely the spaces in (12), and the prescribed steps extend to bimeromorphic diagrams between these families. Near the central fibers, their total spaces are normal, canonical, and $\mathbb{Q}$-Gorenstein. Every nearby fiber is normal and canonical, and is bimeromorphic to the corresponding original fiber X_t. In particular, for every $m \geq 1$ and all sufficiently small t,

$$H^0\left(X_t, \omega_{X_t}^{[m]}\right) \simeq H^0\left(X_t^{(r)}, \omega_{X_t^{(r)}}^{[m]}\right). \tag{13}$$

We first extend the contraction underlying a step. For a flip, we identify the central restriction of the relative canonical algebra, lift finite generation from the prescribed central flip, and take relative Proj. The following local and relative analytic results supply the inputs for these constructions.

(A1) Analytic deformations of canonical singularities. If the central fiber of a flat analytic family over a smooth disk has a canonical singularity at a point, then the relative canonical divisor is rational Cartier near that point and all nearby fibers have canonical singularities there. The total space is canonical, and the exceptional discrepancies of the pair $\left(\mathfrak{X}, X_0\right)$ are nonnegative. The relative $\mathbb{Q}$-Gorenstein property and deformation statements are recorded in [11, Lemma 10]; deformation of canonical singularities is given by [17, VI.5.2]. The total-space and pair statements over a smooth disk are recorded in [11, Remark 10.3]. Finitely many such neighborhoods cover a compact central fiber. Properness then permits a further shrinking of the disk so that their union contains the entire restricted family.

(A2) Deformation of a contraction. Let $\varphi_0 : U_0 \to V_0$ be a proper contraction between compact normal complex spaces. If $(\varphi_0)_*\mathcal{O}_{U_0} = \mathcal{O}_{V_0}$ and $R^1(\varphi_0)_*\mathcal{O}_{U_0} = 0$, then the contraction extends over every given flat analytic

deformation $\mathfrak{U} \to D$ of U_0: after shrinking D, there is a flat analytic deformation $\mathfrak{V} \to D$ of V_0 and a proper morphism $\mathfrak{U} \to \mathfrak{V}$ extending φ_0. This is the proper case of Kollár–Mori's deformation theorem [12, Proposition 11.4, part (11.4.1)].

(A3) Pluricanonical restriction for a projective bimeromorphic morphism. Let $g : Y \to Z$ be a projective bimeromorphic analytic morphism, with Y smooth and Z normal. If $D_Y \subset Y$ is a smooth divisor not contracted by g, then

$$g_*\omega_Y^{\otimes m}(mD_Y) \twoheadrightarrow \left(g|_{D_Y}\right)_*\omega_{D_Y}^{\otimes m} \qquad (m \geq 1). \tag{14}$$

This special case of Nakayama's relative restriction theorem [17, VI.3.8] is stated in [11, Theorem 9]. Projectivity here is relative to Z.

(A4) Lifting graded finite generation near a central compact set. Let $g : Y \to Z$ be a proper morphism of normal complex spaces, let L be a line bundle on Y, and let $W \subset Z$ be a closed analytic subspace. If

$$\mathcal{O}_W \underset{\mathcal{O}_Z}{\otimes} \bigoplus_{m \geq 0} g_*L^{\otimes m}$$

is finitely generated, then every compact subset of W has a neighborhood on which the original graded algebra is finitely generated [11, Lemma 11]. In the application below, finite generation is supplied by the prescribed central flip.

3.3 Relative vanishing and extension of a contraction

Consider one step and suppress its index. The central source X_0 is canonical, hence klt, and $-K_{X_0}$ is φ_0-ample. Fujino's vanishing theorem for projective analytic morphisms [6, Theorem 1.1(ii)] gives

$$R^j(\varphi_0)_*\mathcal{O}_{X_0} = 0 \qquad (j > 0). \tag{15}$$

To apply it, take a projective log resolution $b : W \to X_0$ and write

$$K_W = b^*K_{X_0} + E, \qquad E \geq 0, \qquad A = \lceil E \rceil, \qquad B = A - E.$$

Then B is an SNC boundary with coefficients in $[0, 1)$, and one has the identity of rational line bundles

$$A - (K_W + B) = b^*\big(-K_{X_0}\big).$$

Apply the cited vanishing theorem to $b : W \to X_0$, $\varphi_0 : X_0 \to Z_0$, $L = \mathcal{O}_W(A)$, $H = -K_{X_0}$, and direct-image degree $q = 0$. It gives $R^j(\varphi_0)_*b_*\mathcal{O}_W(A) = 0$. Since A is effective and b-exceptional, normality implies $b_*\mathcal{O}_W(A) = \mathcal{O}_{X_0}$, proving (15).

The central contraction satisfies $(\varphi_0)_*\mathcal{O}_{X_0} = \mathcal{O}_{Z_0}$. By (A2), after shrinking the disk it extends to

$$\varphi : \mathfrak{X} \to \mathfrak{Z}, \qquad h : \mathfrak{Z} \to D, \qquad f = h\varphi,$$

where $\mathfrak{Z} \to D$ is proper and flat with central fiber Z_0, and φ is a proper contraction with connected fibers. Openness of normality in a flat deformation permits a shrinking for which $\mathfrak{Z}$ and its fibers are normal. The stronger canonical property for the source follows from (A1).

The same result makes $K_{\mathfrak{X}/D}$ rational Cartier. Compactness of the central fiber allows us to choose one integer $a > 0$ such that $aK_{\mathfrak{X}/D}$ is Cartier near the central fiber and aK_{X_0} is Cartier. Adjunction on the smooth locus of the central fiber, followed by extension on the normal fiber, gives

$$\mathcal{O}_{\mathfrak{X}}\left(aK_{\mathfrak{X}/D}\right)|_{X_0} \simeq \mathcal{O}_{X_0}\left(aK_{X_0}\right).$$

Thus $\mathcal{O}_{\mathfrak{X}}\left(-aK_{\mathfrak{X}/D}\right)$ restricts to a φ_0-ample line bundle. Relative ampleness for a proper analytic morphism is open on the target. Locally, this can be verified using relative Serre vanishing for sufficiently high powers, the evaluation maps, and the resulting projective embeddings; finitely many target neighborhoods suffice along Z_0. Consequently the line bundle is φ-ample over some neighborhood of Z_0. Since h is proper, we may shrink D so that the entire restricted target lies in this neighborhood. It follows that φ is projective and $-K_{\mathfrak{X}/D}$ is relatively ample.

3.4 Bimeromorphicity on every nearby fiber

Choose $z \in Z_0$ such that the central contraction is an isomorphism near z, and such that both z and its unique preimage x are smooth points of their central fibers. Such points exist because φ_0 is bimeromorphic. A flat morphism is smooth at a smooth point of its fiber, so the two total spaces are smooth near x and z. The vertical differential of φ is invertible, and its induced map on the base is the identity. Its total differential is therefore invertible. The inverse function theorem gives a local isomorphism between neighborhoods of x and z.

After shrinking, this is an isomorphism on the *full preimage* of a target neighborhood. Indeed, choose a sufficiently small source neighborhood U on which the local inverse is defined. The set $\mathfrak{X}\backslash U$ is closed, and properness of φ makes its image closed. That image does not contain z, since $\varphi^{-1}(z) = \{x\}$. Removing this image from the target neighborhood and taking a smaller open subset V gives

$$\varphi^{-1}(V) \xrightarrow{\sim} V. \tag{16}$$

Since h is a submersion at z, the image $h(V)$ contains a neighborhood of the origin. After a further shrinking, (16) meets every fiber.

The central spaces are connected, compact, and normal, and hence have $h^0(\mathcal{O}) = 1$. Upper semicontinuity for proper maps implies that nearby fibers also have $h^0(\mathcal{O}) = 1$, so they are connected. Distinct irreducible components of a normal space are disjoint; the nearby fibers are therefore irreducible. Each $\varphi_t : X_t \to Z_t$ is proper and surjective and contains the nonempty open isomorphism supplied by (16). It is thus bimeromorphic. The total morphism φ is bimeromorphic as well.

If φ_0 is a divisorial contraction, the prescribed central MMP gives canonical singularities on Z_0. Apply (A1) to $h : \mathfrak{Z} \to D$ and set $\mathfrak{X}^+ = \mathfrak{Z}$. This extends the step, with all the asserted properties of the nearby total space, fibers, and relative canonical divisor.

3.5 The flip algebra and central base change

Suppose now that φ_0 is small. After shrinking the disk, the total morphism φ is also small. If a horizontal exceptional prime $F \subset \mathfrak{X}$ existed, properness would force it to meet the central fiber. Its intersection with X_0 would contain a divisor of X_0. Since both F and $\varphi(F)$ dominate the disk, $\varphi(F) \cap Z_0$ would have dimension at most $\dim X_0 - 2$, contradicting the smallness of φ_0. A vertical prime divisor must be an entire irreducible fiber. Since the fiber maps are bimeromorphic, such a divisor cannot be exceptional either. Components outside the central neighborhood can be excluded by shrinking the disk.

Take a projective log resolution $\tau : Y \to \mathfrak{X}$ such that the strict transform D_Y of X_0 is smooth. Set

$$g = \varphi\tau : Y \to \mathfrak{Z}, \qquad L = \omega_Y(D_Y).$$

Then g is projective and bimeromorphic, and D_Y is not contracted by g. The assertion about the central pair in (A1) gives the canonical comparison

$$K_Y + D_Y = \tau^*(K_{\mathfrak{X}} + X_0) + E, \qquad E \geq 0 \quad \text{is} \quad \tau\text{-exceptional.} \tag{17}$$

The local differential argument of Lemma 3.1, now allowing poles of order at most m along X_0, gives

$$\tau_* L^{\otimes m} = \omega_{\mathfrak{X}}^{[m]}(mX_0) \qquad (m \geq 0).$$

Since φ is small, the reflexive sheaves on both sides are determined by their restrictions to the same open set containing all codimension-one points. Hence

$$\mathcal{B}_m := g_* L^{\otimes m} = \omega_{\mathfrak{Z}}^{[m]}(mZ_0), \qquad \mathcal{B} = \bigoplus_{m \geq 0} \mathcal{B}_m. \tag{18}$$

Here $\omega_{\mathfrak{Z}}^{[m]}(mZ_0)$ denotes the reflexive divisorial sheaf on the normal space $\mathfrak{Z}$.

Apply (A3) to g and D_Y. The morphism $D_Y \to X_0$ is a smooth resolution. By Lemma 3.1 and the smallness of φ_0,

$$\left(g|_{D_Y}\right)_* \omega_{D_Y}^{\otimes m} = (\varphi_0)_* \omega_{X_0}^{[m]} = \omega_{Z_0}^{[m]}.$$

We therefore obtain surjections

$$\mathcal{B}_m \twoheadrightarrow \omega_{Z_0}^{[m]} \tag{19}$$

for every $m \geq 1$, compatible with multiplication. In degree zero the map is $\mathcal{O}_{\mathfrak{Z}} \twoheadrightarrow \mathcal{O}_{Z_0}$.

Let t be the disk coordinate. The kernel of (19) is exactly $t\mathcal{B}_m$. At a general smooth point of the central prime divisor Z_0, the total family is smooth and the restriction map is ordinary restriction under adjunction. Thus a local section s with zero restriction is divisible by t there. At the general point of any other prime divisor, t is a unit. The meromorphic section s/t of $\mathcal{B}_m$ is therefore holomorphic at every codimension-one point. Since $\mathcal{B}_m$ is reflexive and $\mathfrak{Z}$ is normal, s/t extends uniquely across the remaining subset of codimension at least two. The reverse inclusion is immediate. Consequently one has an identity of graded algebras

$$\mathcal{B}/t\mathcal{B} = \bigoplus_{m \geq 0} \omega_{Z_0}^{[m]}. \tag{20}$$

The right-hand algebra is finitely generated, and its relative Proj is precisely the *prescribed* central flip X_0^+. Indeed, writing $\varphi_0^+ : X_0^+ \to Z_0$ for the prescribed flip, this morphism is small and $K_{X_0^+}$ is rational Cartier and relatively ample, so its relative canonical algebra agrees with the right-hand side in every degree. Choose $a > 0$ such that $aK_{X_0^+}$ is Cartier and relatively ample. This multiple gives finite generation of the ath Veronese subalgebra. For each $0 \leq r < a$, relative Serre finiteness makes

$$\bigoplus_{q \geq 0} (\varphi_0^+)_* \left(\omega_{X_0^+}^{[r]} \otimes \mathcal{O}_{X_0^+}\left(qaK_{X_0^+}\right)\right)$$

a finite module over that subalgebra. These finitely many residue modules prove finite generation of the whole algebra.

Equation (18) expresses $\mathcal{B}$ as the proper direct-image algebra of the powers of the line bundle L. Applying (A4) together with (20) proves finite generation of $\mathcal{B}$ on a neighborhood of the compact set Z_0. Properness of $\mathfrak{Z} \to D$ permits another shrinking of D, after which we may define over the entire restricted target

$$\mathfrak{X}^+ = \operatorname{Proj}_{\mathfrak{Z}} \mathcal{B}. \tag{21}$$

3.6 Flatness and nearby canonical fibers

By (20), the central fiber of relative Proj is

$$\left(\mathfrak{X}^{+}\right)_{0} = \operatorname{Proj}_{Z_0}\left(\mathcal{B}/t\mathcal{B}\right) = X_0^{+}.$$

Fix $c \in D$ and write $u = t - c$ for a local parameter at c. Since $\mathfrak{Z} \to D$ is flat, multiplication by u is injective on the local rings of $\mathfrak{Z}$ along Z_c. Each $\mathcal{B}_m$ is torsion-free on the normal space $\mathfrak{Z}$, so multiplication by u is also injective on every graded piece. The same holds after homogeneous localization and passage to the degree-zero part, and hence on the standard local algebras of relative Proj. The analytic local rings of these charts are flat over the corresponding local algebras of finite type, so multiplication by u remains injective after analytification. They are therefore torsion-free, and hence flat, over the discrete valuation ring $\mathcal{O}_{D,c}$. As c was arbitrary, $\mathfrak{X}^{+} \to D$ is flat. It is proper because $\mathfrak{X}^{+} \to \mathfrak{Z}$ is projective and $\mathfrak{Z} \to D$ is proper.

The central fiber X_0^{+} is the prescribed canonical space. By (A1), after shrinking the disk the total space $\mathfrak{X}^{+}$ is normal, canonical, and $\mathbb{Q}$-Gorenstein, and all nearby fibers are canonical. On the open set in (16), the canonical sheaf is a line bundle, and (21) is the open set itself. This open set meets every sufficiently nearby fiber. Connectedness and irreducibility follow as before, so $X_t^{+} \to Z_t$ is bimeromorphic for each nearby t. Consequently $X_t \dashrightarrow X_t^{+}$ is bimeromorphic as well.

Since the central morphism $X_0^{+} \to Z_0$ is small, the same argument with horizontal and vertical prime divisors proves that $\mathfrak{X}^{+} \to \mathfrak{Z}$ is small. Choose a sufficiently divisible integer a such that the Proj sheaf $\mathcal{O}(a)$ is a relatively ample line bundle and $aK_{\mathfrak{X}^{+}/D}$ is Cartier. On the common open set containing all codimension-one points, $\mathcal{O}(a)$ agrees with $\omega^{[a]}_{\mathfrak{X}^{+}/D}\left(aX_0^{+}\right)$. Normality extends this isomorphism to the whole space. The central fiber X_0^{+} is the principal Cartier divisor defined by the parameter, so twisting by it does not affect relative ampleness. Hence $K_{\mathfrak{X}^{+}/D}$ is ample relative to $\mathfrak{X}^{+} \to \mathfrak{Z}$. This completes the analytic extension of the prescribed flip.

3.7 Iteration of the finite sequence

At each step, the construction preserves the proper flat family and the prescribed canonical central fiber required for the next step. We may therefore extend all the steps of (12) successively. Since their number r is finite, the intersection of the finitely many successive disk neighborhoods still contains a disk about the origin. The central fiber of the final family is exactly $X_0^{(r)}$. Its total space and nearby fibers have the asserted singularity properties, and every nearby fiber is bimeromorphic to the original fiber through the finite sequence of diagrams. Lemma 3.1 gives (13), completing the proof of Proposition 3.2.

In particular, applying this proposition to the finite sequence of Proposition 2.1 produces, from the original family, a proper flat family

$$f_{\min} : \mathfrak{X}_{\min} \to D. \tag{22}$$

Its central fiber is compact Kähler and terminal, with semiample canonical divisor. Terminality implies that the central fiber is smooth in codimension two, and a projective resolution of a compact Kähler space is again compact Kähler. The total space is canonical and $\mathbb{Q}$-Gorenstein, and the nearby fibers are canonical. The resulting family therefore satisfies the hypotheses of Theorem 4.1.

4 Extension of pluricanonical sections

For a smooth central fiber in Fujiki's class $\mathcal{C}$ and a pluricanonical section with smooth reduced zero divisor, Levine's cyclic-cover method expresses the extension obstruction as the trace of an exterior derivative, whose cohomology class vanishes by Hodge theory [13, Proposition 1.6]. The same cover recovers the lower pluricanonical powers [13, Corollary 1.8]. His semiample-central-fiber result for smooth families with fibers in Fujiki's class $\mathcal{C}$ is the smooth predecessor of the theorem below [13, Corollary 1.10].

For projective central fibers, Kollár–Mori proved the corresponding invariance theorem for log-terminal singularities smooth in codimension two and a globally generated locally free pluricanonical power [12, Theorem 12.5.5]. This includes the projective case of the canonical statement below. For terminal central fibers, Kollár records that the proof also applies in Fujiki's class $\mathcal{C}$ [11, comments following Conjecture 17].

Cao and Păun further develop the infinitesimal method, proving a lifting criterion under a weighted L^2 condition with a Kähler central fiber [2, Theorem 1.1].

Theorem 4.1 establishes this class-$\mathcal{C}$ extension statement for canonical central fibers smooth in codimension two, with semiample canonical divisor and a compact Kähler modification. Its terminal special case suffices for the proof of Theorem 1.1; the canonical formulation gives the additional scope of the extension theorem.

We give an explicit logarithmic period proof based on the cyclic-cover method of Levine and Kollár–Mori. Cohen–Macaulayness of the reflexive canonical powers identifies the relevant first cohomology groups with those of the smooth locus. Logarithmic Hodge theory on a compact Kähler modification of the central cover, combined with a Leray comparison, gives the intrinsic period injection used by the infinitesimal argument.

Theorem 4.1

Let $f : \mathfrak{X} \to \Delta$ be a proper flat holomorphic map with connected central fiber $X = X_0$ of dimension $\dim_{\mathbb{C}} X = n \geq 3$. Suppose that, on a neighborhood of X, the total space $\mathfrak{X}$ is normal, $\mathbb{Q}$-Gorenstein, and has canonical singularities. Assume moreover that:

(i) X is normal, has canonical singularities, and satisfies $\operatorname{codim}_X \operatorname{Sing} X \geq 3$;

(ii) there exists a proper bimeromorphic morphism $b : \widetilde{X} \to X$ from a compact Kähler manifold;

(iii) the canonical $\mathbb{Q}$-line bundle K_X is semiample.

Then, for every fixed integer $m \geq 1$, every section of $H^0\left(X, \omega_X^{[m]}\right)$ extends, after shrinking the disk, to a holomorphic section of

$$\mathcal{F}_m := \mathcal{O}_{\mathfrak{X}}\left(mK_{\mathfrak{X}/\Delta}\right).$$

After shrinking so that the nearby fibers are normal and have canonical singularities, the function

$$t \mapsto h^0\left(X_t, \omega_{X_t}^{[m]}\right)$$

is constant in a neighborhood of 0.

The analytic deformation results in (A1) guarantee the asserted shrinking for the nearby fibers, together with canonical singularities of the total space and the relative $\mathbb{Q}$-Gorenstein property over a smooth disk. A terminal central fiber satisfies (i). A compact Kähler central fiber admits a projective resolution that is again compact Kähler, and hence satisfies (ii). The conclusion is local for each fixed degree m.

Proof

Fix the pluricanonical degree m. We choose a globally generated canonical power of degree $k \geq m$ and a general central section s_0. The character decomposition of its cyclic cover will recover every canonical degree from 1 to k. Given a lift s_r of s_0 to order r, we construct the corresponding Artinian cover and prove a period injection for it and its truncations. Levine's trace formula then annihilates the obstruction to lifting s_0 to order $r+1$. This yields lifts to every finite order; the character summands supply the required degree-m lifts, and coherence gives analytic extension.

4.1 Canonical powers and cohomology on the smooth locus

We first establish a local fact. If Z is $\mathbb{Q}$-Gorenstein with canonical singularities, then $\mathcal{O}_Z(jK_Z)$ is maximal Cohen–Macaulay for *every integer* j, including negative integers. Let a be a local canonical index, and trivialize aK_Z on a sufficiently small neighborhood. The algebra of the corresponding index-one cover is

$$\bigoplus_{b=0}^{a-1} \mathcal{O}_Z(-bK_Z).$$

This finite cover is Gorenstein and has canonical singularities. More explicitly, it is étale in codimension one, and the discrepancies of a divisorial valuation and a lift of that valuation satisfy

$$a\left(E, Z'\right) + 1 = e(a\left(F, Z\right) + 1), \qquad e \geq 1.$$

Thus canonical singularities are preserved under this cover. Canonical singularities are rational and therefore Cohen–Macaulay. A system of parameters downstairs is a system of parameters in each of the finite local rings upstairs, and hence is a regular sequence on the finite direct image. Each character summand displayed above is a direct summand of that module and is therefore Cohen–Macaulay. Tensoring with powers of the invertible sheaf associated with aK_Z proves the assertion for every integer j.

Apply this observation to $\mathfrak{X}$, using the notation $\mathcal{F}_j$ for arbitrary integers j. The coordinate t of the disk is a nonzerodivisor on each torsion-free rank-one sheaf $\mathcal{F}_j$, so $\mathcal{F}_j$ is flat over the disk. Set

$$A_r = \mathbb{C}\left[t\right]/\left(t^{r+1}\right), \qquad X_r = \mathfrak{X} \times_{\Delta} \mathrm{Spec} A_r, \qquad F_{j,r} = \mathcal{F}_j / t^{r+1}\mathcal{F}_j.$$

Then $F_{j,r}$ is a Cohen–Macaulay module of dimension n and is flat over A_r.

Write $\Sigma = \mathrm{Sing} X$ and $U_r = X_r \backslash \Sigma$. At every point of U_0, the flat morphism has a smooth fiber, so f is smooth on a neighborhood of that point. Consequently,

$$F_{j,r}\big|_{U_r} = \omega_{U_r/A_r}^{\otimes j}. \tag{23}$$

In particular, $F_{j,0}$ is a rank-one Cohen–Macaulay module on the normal space X. All of its associated primes are minimal; it is therefore torsion-free, and it satisfies S_2. Thus it is reflexive. The identification on the smooth locus in (23) gives

$$F_{j,0} = \omega_X^{[j]}. \tag{24}$$

For every nearby normal canonical fiber X_c, multiplication by $t - c$ is again injective on $\mathcal{F}_j$. Its quotient is a rank-one Cohen–Macaulay sheaf on X_c, hence reflexive, and agrees with the canonical power on the regular locus. Consequently,

$$\mathcal{F}_j\big|_{X_c} = \omega_{X_c}^{[j]}.$$

The ideal defining Σ has grade at least three on every $F_{j,r}$. Indeed, for a Cohen–Macaulay module, this grade is the infimum of the localized depths at primes containing the ideal. These depths equal the corresponding codimensions in the support, all of which are at least three. The Artinian thickening does not change these codimensions. Local cohomology therefore gives

$$\mathcal{H}_{\Sigma}^{i}\left(F_{j,r}\right) = 0 \qquad (i < 3).$$

The local-to-global spectral sequence for cohomology with supports, together with the long exact sequence for the complement, yields

$$H^1\left(X_r, F_{j,r}\right) \xrightarrow{\sim} H^1\left(U_r, \omega_{U_r/A_r}^{\otimes j}\right). \tag{25}$$

The same grade computation gives the Hartogs extension property in codimension two used below.

Since $\mathfrak{X}$ is $\mathbb{Q}$-Gorenstein and X is compact, there is a common Cartier index on a neighborhood of X. Taking a common multiple with a semiample exponent on the central fiber, choose $k \geq \max\{m, 2\}$ such that $\mathcal{F}_k$ is invertible and $F_{k,0} = \mathcal{O}_X(kK_X)$ is generated by its global sections.

4.2 The central canonical cyclic cover

We choose a section $s_0 \in H^0(X, kK_X)$ whose canonical root cover is normal, Gorenstein, and canonical, and whose inverse image of X_{reg} is smooth. We follow the canonical-cover construction of Kawamata recalled in [12, Lemma 12.5.10], giving the Gorenstein and analytic Bertini details used below.

Consider the analytic canonical line space

$$p : Z = \mathrm{Spec}_X\left(\bigoplus_{j\geq 0}\mathcal{O}_X(-jK_X)\right) \to X. \tag{26}$$

This algebra is locally of finite type: after trivializing aK_X, its degree-$j+a$ part is obtained from its degree-j part by multiplication with a local generator in degree a. Over X_{reg}, the space Z is the usual total space of K_X.

Locally, take the index-one cover $\widehat{X} \to X$ with group μ_a, and choose a canonical generator θ with character χ. Then

$$Z \simeq (\widehat{X} \times \mathbb{C}_z)/\mu_a, \qquad g(x, z) = (gx, \chi(g)^{-1}z). \tag{27}$$

Indeed, the coefficient of z^j in an invariant polynomial has character χ^j, which gives precisely $\mathcal{O}_X(-jK_X)$. The top form $\theta \wedge dz$ is invariant. The finite quotient is normal and Cohen–Macaulay, and this invariant generator trivializes the dualizing sheaf on the codimension-one locus. Reflexive extension identifies that sheaf with an invertible trivial sheaf, so Z is Gorenstein. The quotient is étale in codimension one, and $\widehat{X} \times \mathbb{C}$ has canonical singularities. The discrepancy formula implies that Z is klt. Since K_Z is Cartier, its exceptional discrepancies are integers, hence nonnegative; thus Z is canonical.

Let $W = H^0(X, kK_X)$, and denote by τ_k the degree-k tautological section of $p^*(kK_X)$. The finite-dimensional system

$$\mathbb{C}\tau_k + p^*W \tag{28}$$

is basepoint-free on all of Z: at a point lying over x, some section of W is nonzero at x. Take a log resolution of Z and write

$$K_{\widetilde{Z}} = g^{*}K_{Z} + \sum_{i} a_{i}E_{i}, \qquad a_{i} \geq 0.$$

The pullback of (28) remains basepoint-free. A general member is smooth on $\widetilde{Z}$, is transverse to every exceptional stratum, and contains no exceptional center. These assertions follow by applying Sard's theorem to the projection from the smooth incidence space and then applying the same argument on each stratum. Even if $\widetilde{Z}$ is noncompact, there are at most countably many strata; the bad parameter values form a countable union of measure-zero sets and can therefore be avoided simultaneously. We may also avoid the hyperplane on which the coefficient of τ_{k} vanishes, and arrange that the zero divisor of the resulting s_{0} is smooth and reduced on X_{reg}.

After normalizing the coefficient of τ_{k}, the member has the form

$$Y_{0} = \{\tau_{k} - p^{*}s_{0} = 0\}. \tag{29}$$

Its total transform on the resolution has no exceptional divisorial component. As a Cartier hypersurface in the Cohen–Macaulay space Z, the space Y_{0} is Cohen–Macaulay. Bertini's theorem, together with the requirement that no component of $\mathrm{Sing}Z$ be contained in Y_{0}, shows that it is smooth in codimension one. Hence Y_{0} is normal. Adjunction on its smooth strict transform gives the nonnegative exceptional coefficients a_{i} above, so Y_{0} is canonical. It is also Gorenstein, being a Cartier hypersurface in a Gorenstein space.

The monic relation of degree k defines the finite cover

$$q_{0} : Y_{0} \to X, \qquad (q_{0})_{*}\mathcal{O}_{Y_{0}} = \bigoplus_{a=0}^{k-1} \mathcal{O}_{X}(-aK_{X}). \tag{30}$$

Over X_{reg}, its equation is $w^{k} = s_{0}$. Since the branch divisor is smooth and reduced, $q_{0}^{-1}\left(X_{\mathrm{reg}}\right)$ is smooth, including at the ramification points. Set

$$Z_{0} = q_{0}^{-1}\Sigma, \qquad V_{0} = Y_{0} \backslash Z_{0}.$$

Finiteness gives $\mathrm{codim}_{Y_{0}} Z_{0} \geq 3$.

4.3 Cyclic covers over Artinian thickenings

Suppose that a section $s_{r} \in H^{0}\left(X_{r}, F_{k,r}\right)$ has already been constructed, with central restriction equal to the section s_{0} chosen above. Define

$$q_{r} : Y_{r} = \mathrm{Spec}_{X_{r}}\left(\bigoplus_{a=0}^{k-1} F_{-a,r}\right) \to X_{r}. \tag{31}$$

Multiplication in degrees reaching k is reduced using s_r. The multiplication maps are induced by the canonical-power multiplication on the total space and the invertibility of $\mathcal{F}_k$, and remain defined after reduction modulo t^{r+1}. On U_r, this is the usual algebra with equation $w^k = s_r$. Associativity may also be checked on that open set and then extended by the Hartogs property of the target modules. Thus (31) is a well-defined global algebra.

Each summand is flat over A_r and maximal Cohen–Macaulay. These properties of the finite direct image imply that Y_r is flat over A_r and Cohen–Macaulay. Let $V_r = Y_r \backslash Z_0$, identifying the underlying topological spaces of the thickenings with their central fibers. Then V_r/A_r is smooth. Away from the branch locus, kw^{k-1} is a unit. At a branch point, a fiber-coordinate derivative of s_0 is a unit and remains a unit on the thickening. Replacing that coordinate by s_r puts the equation into the form $z_1 = w^k$.

Set

$$L_r = q_r^* F_{k,r}. \tag{32}$$

This is an invertible sheaf. The total space of the line bundle ω_{U_r/A_r} has trivial canonical bundle relative to A_r: the canonical factor from the base cancels the inverse factor from the line-bundle direction. Adjunction for the degree-k equation gives

$$L_r|_{V_r} = \omega_{V_r/A_r}. \tag{33}$$

All these identifications commute with truncation. The ideal of the removed set has grade at least three on the invertible sheaf L_r on the Cohen–Macaulay space Y_r. Hence

$$H^1\left(Y_r, L_r\right) \xrightarrow{\sim} H^1\left(V_r, \omega_{V_r/A_r}\right). \tag{34}$$

Since $F_{k,r}$ is invertible, the projection formula gives

$$(q_r)_* L_r = \bigoplus_{a=0}^{k-1} F_{k-a,r} = \bigoplus_{\ell=1}^{k} F_{\ell,r}. \tag{35}$$

For each existing s_r, (31) thus defines a cover over A_r, compatibly with its truncations.

4.4 The logarithmic period injection

We first construct a compact Kähler resolution of Y_0. Choose the compact Kähler resolution $b : \widetilde{X} \to X$ in hypothesis (ii). Since the canonical discrepancies are nonnegative, the pullback of s_0 can be regarded as a holomorphic section of $kK_{\widetilde{X}}$. Its root hypersurface is a compact subspace of $\mathbf{P}\left(\mathcal{O}_{\widetilde{X}} \oplus K_{\widetilde{X}}\right)$; the monic equation ensures that it does not meet the divisor at infinity. Its algebra is locally free over $\widetilde{X}$ and generically separable, so the hypersurface is reduced. A projective embedded resolution is compact Kähler. Over the dense open set where b is an isomorphism, this root cover agrees with Y_0.
The resulting map to the normal finite cover Y_0 extends over the entire resolution. Indeed, locally choose integral generators for the finite algebra defining Y_0 over the base. On the dense open set where the map is defined, their pullbacks satisfy monic equations with locally bounded coefficients, and the roots are consequently locally bounded. The removable-singularity theorem on a normal space extends them holomorphically. Their defining equations and gluing identities continue from the dense open set. The resulting map is proper and bimeromorphic on each component. Further projective blowups principalize the inverse-image ideal of Z_0 and give

$$\tilde{q} : \widetilde{Y} \to Y_0, \qquad E = \left(\tilde{q}^{-1} Z_0\right)_{\mathrm{red}},$$

where $\widetilde{Y}$ is compact Kähler and E is a simple normal crossing divisor. Set

$$W = \widetilde{Y} \backslash E, \qquad h = \tilde{q}|_W : W \to V_0.$$

The morphism h is a proper modification between smooth spaces.
Since Y_0 is normal and Gorenstein, (33) extends reflexively to $L_0 = \omega_{Y_0}$. Canonical singularities give inclusions

$$\tilde{q}^* L_0 \hookrightarrow \omega_{\widetilde{Y}} \hookrightarrow \Omega^n_{\widetilde{Y}}(\log E).$$

We claim that the induced map

$$H^1\left(Y_0, L_0\right) \to H^1\left(\widetilde{Y}, \Omega^n_{\widetilde{Y}}(\log E)\right)$$

is injective. The sheaf identity in Lemma 3.1 gives $h_* \omega_W = \omega_{V_0}$. The low-degree Leray sequence therefore gives an injection

$$H^1\left(V_0, \omega_{V_0}\right) \hookrightarrow H^1\left(W, \omega_W\right).$$

It is induced by pullback of top differentials. Restricting the claimed map to W gives the composite of this injection with the isomorphism (34) for $r = 0$. This proves the claim.

For the compact Kähler simple normal crossing pair $\left(\widetilde{Y}, E\right)$, we use logarithmic Hodge-to-de Rham degeneration in the compact Kähler formulation of Liu, Rao, and Wan [15, Theorem 3.2]. The spectral sequence degenerates at E_1. Since there are no logarithmic forms of degree greater than n, the edge map from the top column,

$$H^1(\widetilde{Y}, \Omega^n_{\widetilde{Y}}(\log E)) \to \mathbb{H}^{n+1}(\widetilde{Y}, \Omega^\bullet_{\widetilde{Y}}(\log E)),$$

is injective. Logarithmic de Rham comparison [15, Section 3.2] identifies the target with $H^{n+1}(W, \mathbb{C})$. We have therefore obtained an injection from $H^1(Y_0, L_0)$ to $H^{n+1}(W, \mathbb{C})$.

Define per_0 by restriction to V_0, inclusion of its top holomorphic de Rham term, and holomorphic de Rham comparison. Naturality under restriction and pullback identifies the preceding injective composite with

$$H^1(Y_0, L_0) \xrightarrow{\mathrm{per}_0} H^{n+1}(V_0, \mathbb{C}) \xrightarrow{h^*} H^{n+1}(W, \mathbb{C}).$$

Consequently, the central period map is injective:

$$\mathrm{per}_0 : H^1(Y_0, L_0) \hookrightarrow H^{n+1}(V_0, \mathbb{C}). \tag{36}$$

4.5 Period injections over Artinian thickenings

Fix an existing lift s_r and its compatible truncations. For $0 \le j \le r$, the morphism V_j/A_j is smooth. The relative holomorphic Poincaré lemma gives a canonical quasi-isomorphism

$$A_j \simeq \Omega^\bullet_{V_j/A_j}.$$

In local smooth coordinates this is the ordinary Poincaré lemma applied coefficient by coefficient in t. The canonical inclusions of the constant sheaf identify these local quasi-isomorphisms on overlaps. Consequently,

$$\mathbb{H}^{n+1}\left(V_j, \Omega^\bullet_{V_j/A_j}\right) = H^{n+1}(V_0, \mathbb{C}) \underset{\mathbb{C}}{\otimes} A_j.$$

Restriction to V_j, followed by inclusion of the top de Rham term, defines

$$\mathrm{per}_j : H^1\left(Y_j, L_j\right) \to H^{n+1}(V_0, \mathbb{C}) \underset{\mathbb{C}}{\otimes} A_j. \tag{37}$$

These maps commute both with reduction modulo t and with multiplication by t. We prove by induction that per_j is injective and that

$$H^0\left(Y_j, L_j\right) \to H^0(Y_0, L_0) \quad \text{is surjective.} \tag{38}$$

The initial period injection is (36), and the assertion about sections is automatic for $j = 0$. For $j \geq 1$, regard L_{j-1} and L_0 as sheaves on Y_j through their natural closed embeddings. Flatness gives an exact sequence

$$0 \to L_{j-1} \xrightarrow{t} L_j \to L_0 \to 0. \tag{39}$$

Assume that per_{j-1} is injective. If $b \in H^1\left(Y_{j-1}, L_{j-1}\right)$ maps to zero under multiplication by t, compatibility gives $t\,\mathrm{per}_{j-1}(b) = 0$. Multiplication $t : A_{j-1} \to A_j$ is injective, so $\mathrm{per}_{j-1}(b) = 0$, and hence $b = 0$. Thus the multiplication-by-t map on H^1 in (39) is injective. The long exact sequence gives (38).

Now let $a \in H^1\left(Y_j, L_j\right)$ have zero period. Its reduction to the central fiber vanishes by (36). Hence $a = tb$ for some $b \in H^1\left(Y_{j-1}, L_{j-1}\right)$. The identity $0 = \mathrm{per}_j(a) = t\,\mathrm{per}_{j-1}(b)$ and the same injectivity imply $b = 0$. This proves the injectivity of per_j and completes the induction.

In particular,

$$\mathrm{d} : H^1\left(V_j, \Omega^{n-1}_{V_j/A_j}\right) \to H^1\left(V_j, \omega_{V_j/A_j}\right) \quad \text{is zero.} \tag{40}$$

Indeed, applying d to a Čech cocycle gives a boundary in the total Čech–de Rham complex, so its de Rham period is zero. By (34), this image comes from a unique class in $H^1\left(Y_j, L_j\right)$. The injectivity of (37) then makes it zero.

4.6 Levine’s obstruction and trace formulas

We apply Levine’s local obstruction calculation [13, Lemmas 1.2–1.5, pp. 295–298]. We give the formulas in the present notation, including the identification of the central-section obstruction and the holomorphic map across the ramification divisor.

Suppose again that s_r has been constructed. Work on the smooth open part of the original family, with fiber coordinates $\left(z_i, t\right)$ and transition functions

$$z_i = g_{ij}\left(z_j, t\right), \qquad J_{ij} = \det\left(\frac{\partial\, g_{ij}}{\partial\, z_j}\right), \qquad \rho_{ij} = \sum_{\alpha} \frac{\partial\, g_{ij}^{\alpha}}{\partial\, t}\,\frac{\partial}{\partial\, z_i^{\alpha}}\,.$$

Write the section as $s_i\left(dz_i\right)^k$. Its transition rule is

$$s_i\left(g_{ij}, t\right) = J_{ij}^{-k} s_j \pmod{t^{r+1}}.$$

Differentiating the transition rule gives the cocycle, expressed in the i-coordinates,

$$D(s_r)_{ij} = \left(k \operatorname{div}\left(\rho_{ij}\right) s_i + \rho_{ij}\left(s_i\right)\right)\left(dz_i\right)^k. \tag{41}$$

Let $\delta_{r+1}\left(s_0\right)$ be the connecting class associated with

$$0 \to F_{k,r} \xrightarrow{t} F_{k,r+1} \to F_{k,0} \to 0. \tag{42}$$

Then

$$\left[D\left(s_r\right)\right] = (r+1)\delta_{r+1}\left(s_0\right)\big|_{U_r}. \tag{43}$$

Choose arbitrary next-order lifts of the local coefficients of s_r and adopt the Čech convention

$$J_{ij}^k s_i\left(g_{ij}, t\right) - s_j = t^{r+1}\zeta_{ij} \pmod{t^{r+2}}$$

for the Čech difference. Dividing this difference by t gives the representative $t^r\zeta_{ij}$ of the connecting class in (42). Differentiation by ∂_t gives $(r+1)t^r\zeta_{ij}$ on the right. On the left, the differences of the local $\partial_t s_i$ form a coboundary, and the remaining term is (41), using $\partial_t J_{ij}/J_{ij} = \operatorname{div}\left(\rho_{ij}\right)$. This proves (43). The class in (43) is the connecting image of the central section s_0; the chosen finite-order lift s_r is used to compute it.

On V_r, write the tautological pulled-back canonical section as $\overline{\omega} = w\,dz$, and set

$$\mu_r = \langle \rho, \overline{\omega} \rangle \in H^1\left(V_r, \Omega^{n-1}_{V_r/A_r}\right).$$

Differentiating the transition identity $g_{i\ell} = g_{ij} \circ g_{j\ell}$ with respect to t gives the Čech cocycle identity for ρ. The root coordinates satisfy $w_i\,dz_i = w_j\,dz_j$, so $\overline{\omega}$ is a global section. The contraction therefore defines a cohomology class of $(n-1)$-forms. The contraction is first taken in the pullbacks from U_r of the relative tangent sheaf and canonical line, and is then composed with the natural differential map to forms on V_r. The local equation gives

$$k w^{k-1} \mathrm{d}w = \mathrm{d}s_i.$$

Off the ramification locus, the differential identifies top forms with the pullback of the canonical line of U_r. Under this identification, multiplication by $\overline{\omega}^{k-1}$ defines a map to $q_r^*\left(\omega^{\otimes k}_{U_r/A_r}\right)$. This map extends holomorphically across the ramification locus, as the following local calculation shows. Choose a coordinate index for which $\partial s_i/\partial z_1$ is a unit. A local frame of the canonical line of the cover is

$$\frac{dz}{k w^{k-1}} = \left(\frac{\partial s_i}{\partial z_1}\right)^{-1} \mathrm{d}w \wedge dz_2 \wedge \cdots \wedge dz_n. \tag{44}$$

Multiplication by $(w\,dz)^{k-1}$ sends this frame to $(dz)^k/k$. Thus the operation is a holomorphic sheaf map, including at the ramification locus.

Expanding $\mathrm{d}\left(w\,\iota_\rho dz\right) = \mathrm{d}w\wedge\iota_\rho dz + w\,\mathrm{div}\,(\rho)\,dz$ and using the root equation gives

$$\overline{\omega}^{k-1}\mathrm{d}\mu_r = \left(\mathrm{div}\,(\rho)s_i + \frac{1}{k}\rho\,(s_i)\right)(dz_i)^k. \tag{45}$$

This expression is invariant under the cyclic group. The monic finite cover has total degree k, so its trace is k times this invariant expression. Consequently,

$$\left[D\,(s_r)\right] = \mathrm{Tr}_{V_r/U_r}\left(\overline{\omega}^{k-1}\mathrm{d}\mu_r\right). \tag{46}$$

The same computation applies if the cover is disconnected: the monic algebra still has total rank k. These calculations use only smooth coordinate charts and apply on the smooth open set considered here in every dimension.

By (40), the cohomology class of $\mathrm{d}\mu_r$ vanishes. The holomorphic sheaf map and the trace then make (46) zero. Combining (43) with (25), and using that $r+1$ is invertible over $\mathbb{C}$, gives $\delta_{r+1}\,(s_0) = 0$ in $H^1\left(X_r, F_{k,r}\right)$. Thus s_0 has at least one lift to order $r+1$. Starting with $r=0$, induction therefore shows that, for every $r \geq 0$, there exists $s_r \in H^0\left(X_r, F_{k,r}\right)$ whose restriction to X is s_0.

4.7 Recovery of the lower pluricanonical powers

The recovery of lower powers is the cyclic-cover argument of [13, Corollary 1.8], whose proof Levine credits to Shepherd-Barron. Here we apply the character decomposition over each fixed Artinian base.

Fix a target order r, and choose one lift s_r constructed above together with its own truncations. Apply (38) to this finite chain and use (35). It follows that

$$\bigoplus_{\ell=1}^{k} H^0\left(X_r, F_{\ell,r}\right) \to \bigoplus_{\ell=1}^{k} H^0\left(X, F_{\ell,0}\right) \quad \text{is surjective.}$$

The restriction map respects the direct-sum decomposition. Since $m \leq k$, every section of $H^0\left(X, F_{m,0}\right)$ therefore lifts to order r. Since the target order r was arbitrary, this proves surjectivity at every finite order.

4.8 Passage from finite-order lifts to analytic sections

The passage from lifting to every finite order to analytic local constancy is already present, for line bundles in smooth families, in [13, Lemma 1.1]. For coherent analytic sheaves flat over the base of a proper flat family over a smooth Stein curve, the relation between extension, base change, and torsion freeness is developed in [5, Theorem 1.2 and Appendix A]. We give a direct finite-module argument for the reflexive canonical powers, using the finite-order lifts constructed above.

Set

$$R = \mathbb{C}\{t\}, \qquad E = \left(f_*\mathcal{F}_m\right)_0, \qquad G = \left(R^1 f_*\mathcal{F}_m\right)_0.$$

Proper coherence makes E and G finite R-modules. The bounded exponent of the torsion in G will make one sufficiently high finite-order lift enough for analytic extension. Apply direct image to the exact sequence given by multiplication by t^a on $\mathcal{F}_m$, and take the stalk at 0. The result is

$$0 \to E/t^a E \to H^0\left(X_{a-1}, F_{m,a-1}\right) \xrightarrow{\partial_a} G\left[t^a\right] \to 0, \tag{47}$$

where $G\left[t^a\right] = \ker\left(t^a : G \to G\right)$. In the map from the sequence modulo t^a to the sequence modulo t, the middle map on $\mathcal{F}_m$ is the identity and the left map on $\mathcal{F}_m$ is multiplication by t^{a-1}. Naturality gives

$$\partial_1 \circ \mathrm{res} = t^{a-1}\partial_a. \tag{48}$$

The torsion submodule of the finite R-module G is annihilated by some fixed power t^B. Given a central section, choose $a > B$ and use the lift to order $a-1$ supplied by the preceding step. Its image under ∂_a lies in $G\left[t^a\right]$, hence in the torsion submodule. Equation (48) therefore makes the ∂_1-image of the central section zero. By (47), the section comes from E/tE, and consequently from a holomorphic section germ on the original analytic family. Choosing a basis of the finite-dimensional central section space gives a common smaller disk on which all its elements extend.

Since $\mathcal{F}_m$ is flat over the disk, its direct image $f_*\mathcal{F}_m$ is torsion-free and hence locally free. The surjectivity just proved identifies its central fiber with the entire space $H^0\left(X, F_{m,0}\right)$. At every nearby point, the analogous exact sequence injects the fiber of this locally free direct image into the corresponding fiber cohomology space H^0. This gives the central dimension as a lower bound. Proper upper semicontinuity gives the central dimension as an upper bound for nearby fibers. The two bounds agree, proving local constancy of h^0. Finally, (24) and its counterpart on nearby fibers identify these dimensions with the required reflexive plurigenera.

□

Remark

The codimension-three condition gives the restriction isomorphisms (25) and (34). The compact Kähler resolution is used to establish the central period injection; the Artinian period maps are then constructed on the smooth open thickenings V_j/A_j. Normality and canonical singularities identify the reflexive pluricanonical powers on nearby fibers.

5 Proof of the main theorem

5.1 Fibers of dimension at most two

If the fibers have dimension zero, connectedness makes each of them a single point. Hence $P_m(X_t) = 1$ for every $m \geq 1$.

For one-dimensional fibers, Ehresmann's theorem makes the genus g locally constant. Riemann–Roch and Serre duality give, for a smooth compact connected curve C of genus g,

$$P_m(C) = \begin{cases} 0, & g = 0, \\ 1, & g = 1, \\ g, & g \geq 2 \quad \text{and} \quad m = 1, \\ (2m-1)(g-1), & g \geq 2 \quad \text{and} \quad m \geq 2. \end{cases}$$

Indeed, K_C has negative degree when $g = 0$ and is trivial when $g = 1$. For $g \geq 2$ and $m \geq 2$, the line bundle $K_C^{\otimes(1-m)}$ has negative degree, so its space of sections vanishes. The displayed formula proves local constancy of every plurigenus.

For two-dimensional fibers, the assertion is Iitaka's deformation invariance theorem for compact complex surfaces [10, Theorem III, p. 247]. Since Δ is connected, these local constancy statements prove Theorem 1.1 in dimensions at most two.

5.2 Fibers of dimension at least three

Assume now that $n \geq 3$. Fix an integer $m \geq 1$ and a point $t_0 \in \Delta$. After translating the parameter and restricting to a smaller disk, write $t_0 = 0$. If $P_m(X_0) = 0$, upper semicontinuity for the line bundle $K_{\mathcal{X}/\Delta}^{\otimes m}$ gives $P_m(X_t) = 0$ for all sufficiently small t.

Suppose that $P_m(X_0) > 0$. A nonzero m-canonical section defines an effective divisor D_s with $mK_{X_0} \sim D_s$. The positive current $[D_s]/m$ shows that K_{X_0} is pseudoeffective. Proposition 2.1 therefore supplies a finite ordinary K-MMP

$$X_0 = X_0^{(0)} \dashrightarrow X_0^{(1)} \dashrightarrow \cdots \dashrightarrow X_0^{(N)} = \ : M_0 \tag{49}$$

whose models are compact Kähler and terminal, and whose endpoint has semiample canonical divisor.

Proposition 3.2 extends (49) over a smaller disk $D \ni 0$. Write the resulting family as

$$f : \mathfrak{M} \to D.$$

It is proper and flat, with central fiber M_0. After shrinking D, its total space near the central fiber is normal, canonical, and $\mathbb{Q}$-Gorenstein, and each fiber M_t is normal canonical and bimeromorphic to X_t.

Terminal singularities are smooth in codimension two, so

$$\operatorname{codim}_{M_0} \operatorname{Sing} M_0 \geq 3.$$

A projective resolution of the compact Kähler space M_0 is compact Kähler. Theorem 4.1 therefore gives

$$h^0\left(M_t, \omega_{M_t}^{[m]}\right) = h^0\left(M_0, \omega_{M_0}^{[m]}\right)$$

for sufficiently small t. By Lemma 3.1,

$$\begin{aligned} P_m(X_t) &= h^0\left(M_t, \omega_{M_t}^{[m]}\right) \\ &= h^0\left(M_0, \omega_{M_0}^{[m]}\right) = P_m(X_0). \end{aligned}$$

Thus P_m is locally constant on Δ for every $m \geq 1$. Since the disk is connected, Theorem 1.1 follows.

□

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All mathematical derivations and proofs were produced by AI. The human author formulated the problem, provided relevant literature, and guided the research direction.

References

[1] S. Boucksom, *Divisorial Zariski decompositions on compact complex manifolds*, Ann. Sci. Éc. Norm. Supér. (4) **37** (2004), no. 1, 45–76. doi:10.1016/j.ansens.2003.04.002.

[2] J. Cao and M. Păun, *Infinitesimal extension of pluricanonical forms*, preprint, 9 December 2020, revised 28 March 2023. arXiv:2012.05063v5.

[3] J. Chen, S. Rao, and K. Wang, *Matsumura's extension problem for pluricanonical forms in Kähler families I: the smooth and essentially Moishezon cases*, preprint, 30 September 2026. arXiv:2609.40040v1.

[4] F. Deng, J. Ning, Z. Wang, and X. Zhou, *Invariance of plurigenera for Kähler families with nef canonical bundles*, preprint, 29 September 2026. arXiv:2609.37804v1.

[5] J. Feng, Y. Li, and S. Rao, *Aspects of deformation invariance of plurigenera of Kähler varieties*, preprint, 13 October 2026. arXiv:2610.17817v1.

[6] O. Fujino, *Vanishing theorems for projective morphisms between complex analytic spaces*, Math. Res. Lett. **32** (2025), no. 3, 739–769. Author's version.

[7] C. D. Hacon, Y. Li, and S. Rao, *On Pseudo-Effectivity and Volumes of Adjoint Classes in Kähler Families with Projective Central Fiber*, preprint, 4 February 2026, revised 5 March 2026. arXiv:2602.04158v2.

[8] C. Hacon and L. Xie, *On the Kähler MMP and the transcendental base-point-free theorem*, preprint, 27 July 2026. arXiv:2607.24986v1.

[9] H. Hironaka, *Flattening theorem in complex-analytic geometry*, Amer. J. Math. **97** (1975), no. 2, 503–547. doi:10.2307/2373721.

[10] S. Iitaka, *Deformations of compact complex surfaces II*, J. Math. Soc. Japan **22** (1970), no. 2, 247–261. doi:10.2969/jmsj/02220247.

[11] J. Kollár, *Deformations of varieties of general type*, Milan J. Math. **89** (2021), 345–354. arXiv:2101.10986v3.

[12] J. Kollár and S. Mori, *Classification of three-dimensional flips*, J. Amer. Math. Soc. **5** (1992), no. 3, 533–703. doi:10.1090/S0894-0347-1992-1149195-9.

[13] M. Levine, *Pluri-canonical divisors on Kähler manifolds*, Invent. Math. **74** (1983), 293–303. doi:10.1007/BF01394318.

[14] Y. Li, M. Păun, Z. Wang, and X. Zhou, *Psh variation of singular Kähler–Einstein metrics and Invariance of plurigenera for smooth Kähler families*, preprint, 6 October 2026. arXiv:2610.07826v1.

[15] K. Liu, S. Rao, and X. Wan, *Geometry of logarithmic forms and deformations of complex structures*, J. Algebraic Geom. **28** (2019), 773–815. doi:10.1090/jag/723.

[16] K. Liu and X. Wan, *Invariance of logarithmic plurigenera and adjoint volumes in Kähler families*, preprint, 19 October 2026. arXiv:2610.26519v1.

[17] N. Nakayama, *Zariski-decomposition and abundance*, MSJ Memoirs, vol. 14, Mathematical Society of Japan, Tokyo, 2004.

[18] OpenAI, *Orbifold and logarithmic Iitaka subadditivity*, OAI 033, preprint, 26 September 2026.

[19] OpenAI, *Log abundance for compact Kähler spaces under logarithmic Iitaka subadditivity*, OAI 034, preprint, 4 October 2026.

[20] Y.-T. Siu, *Invariance of plurigenera*, Invent. Math. **134** (1998), 661–673. doi:10.1007/s002220050276.

[21] Y.-T. Siu, *Extension of twisted pluricanonical sections with plurisubharmonic weight and invariance of semipositively twisted plurigenera for manifolds not necessarily of general type*, in *Complex Geometry* (Göttingen, 2000), Springer, Berlin, 2002, pp. 223–277. doi:10.1007/978-3-642-56202-0_15.